

BUSINESS CONTINUITY, BACKUP AND RECOVERY

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1. INTRODUCTION

Business continuity is very important to any data centre. Business continuity defines the processes and procedures that an enterprise undertakes to overcome any planned or unplanned downtime. The downtime can be due to machine problems or due to natural disasters. Such downtime can hamper business operations. To achieve business continuity, a business should plan such that it can continue to operate during and after the disaster (man-made or natural) occurs. There are various measures that can be taken to ensure business continuity. Some techniques include evaluating backup technologies and implementing a successful backup and recovery solution.

Maintaining backups of data ensures that a copy of the data is available even if the original data is damaged or not available due to system outage. The data centre design should include robust solution to facilitate easy recovery and retrieval of data from backups and archives, as needed by the business.

2. BUSINESS CONTINUITY (BC) INFRASTRUCTURE SERVICES

The BC infrastructure services provide support for the highest levels of application availability. These services ensure data availability through the use of networking technology in the following areas:

- **Site Selection:** It refers to functions that enable the automatic detection of failure on the application level. On detection of failure, these functions reroute all subsequent data requests to an alternate available site and server farm. The site selection can be over the internet or the intranet. Some mechanisms for site selection include:

Use of Domain Name Service (DNS)

Host routes

Routed network

- **SAN Extension:** It Refers to extending an existing SAN to a secondary location. This secondary location can be located within the same data centre or it can be on a different geographical location. The extension can be done by enabling data replication from primary to secondary SAN. There are two methods to replicate the data:

- **Synchronous Replication:** Data is written to the primary storage and replica simultaneously. Therefore, primary copy and the replica is always synchronised.

- **Asynchronous Replication:** Data is first written to the primary storage and then copied to the replica. A schedule is followed to copy the data to the replica. This may result in the replica not having a real-time copy of the data.

The selection of the replication technique depends on the following factors:

- Distance between data centres.

- Application transaction rate.

- Latency (data transport speed) between the distributed data centres.

- **Data Centre Interconnectivity:** Data centre interconnectivity services are connectivity alternatives provided by various technologies. These connectivity alternatives support the communication requirements for site selection and SAN extension.

3. NEED FOR REDUNDANCY

Redundancy is used to avoid service failures and data loss. The redundant data is a duplicate data. To reduce the occurrence of data loss, organisations create redundant copies.

e.g.: In the event, if a virtual machine fails, then its copy comes online to reduce the service interruption. The redundant systems are better to use in scenarios where the organisation needs to keep the critical systems online 24x7. In a data centre, redundancy is needed to counter the following failure scenarios:

- **Component failure:** Due to hardware problems or failure in power supply to devices

- **Assembly failure:** Failure in storage and complete server

- **Room failure:** Breakdown in power supply

- Building failure: Due to flames or deluge
- Site failure: Due to break in connectivity throughout the cable
- City failure: Due to failure in power grid
- Regional failure: Due to natural disasters
- Country failure: Due to wars

4. INFORMATION AVAILABILITY

Information availability refers to the capability of the infrastructure to function as per the needs of the business during its period of operations. Information availability means that authorised people should be able to access the data resources when they need it

- Reliability

For any system, there should be no failure under any circumstances for the specified time

- Accessibility

Any information, which is required should be accessible to the right person at right time

- Timeliness

Any information, which is required should be accessed within the given time limit

Note that while defining the accessibility of the information, it is important to consider the following factors:

- System uptime: This is a time period between which the system is accessible.
- System downtime: This is a time period in which the system cannot be accessed.

5. BC TERMINOLOGY

There are some key terms related to BC that you should be aware of to understand the BC environment:

- Disaster recovery
- Disaster restart

- Recovery-Point Objective (RPO)
- Recovery-Time Objective (RTO)

Disaster recovery: This refers to the process of restoring systems, data and the infrastructure that are needed to support the critical business operations in the event of a disaster. It involves restoring a previous copy of the data. After the recovery procedures are completed, the data is validated to verify if it is correct.

Disaster restart: This refers to the process of restarting the business operations with the replicated and consistent copies of data and applications.

Recovery-Point Objective (RPO): This refers to the point in time to which the system and data should be recovered after a failure. It also specifies the amount of data loss that a business is able to sustain in the event of a failure. Greater the RPO, the more tolerant the process is to interruption. Based on the RPO, businesses should define the minimum frequency after which the data backup should be made.

e.g. 1: The backup or replicas should be prepared at least for once in 10 hours, if the particular RPO is for seven hours.

e.g. 2: If the RPO is 1 day, for each midnight, the backups are created on an offsite tape drive. The recovery strategy in this case is to recover data from the last set of backup tapes.

e.g. 3: If the RPO is 1 hour, then in every one hour, the database sends the data logs to a remote site. The recovery strategy is to recover the data at the point of the last log sent.

e.g. 4: If the RPO is zero, it replicates mission-critical data simultaneously to a remote site.

Figure shows an example of the RPO strategy. Here, the recovery strategy is defined based on the recovery point. If the RPO is defined in seconds, then synchronous replication happens. In case of RPO in minutes, then asynchronous replication can be specified. In case of RPO in hours and days, periodic replication can take place. For RPO defined in weeks, the tape backup can be done.

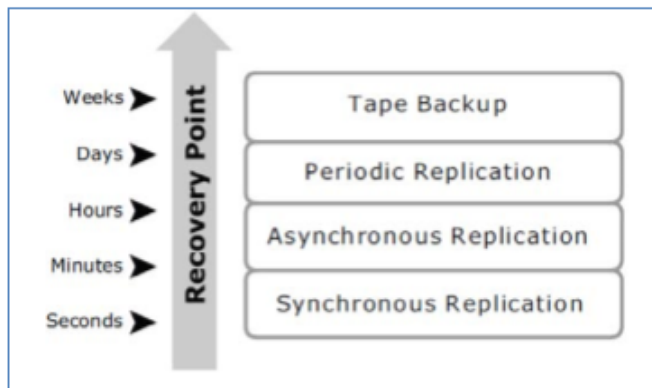


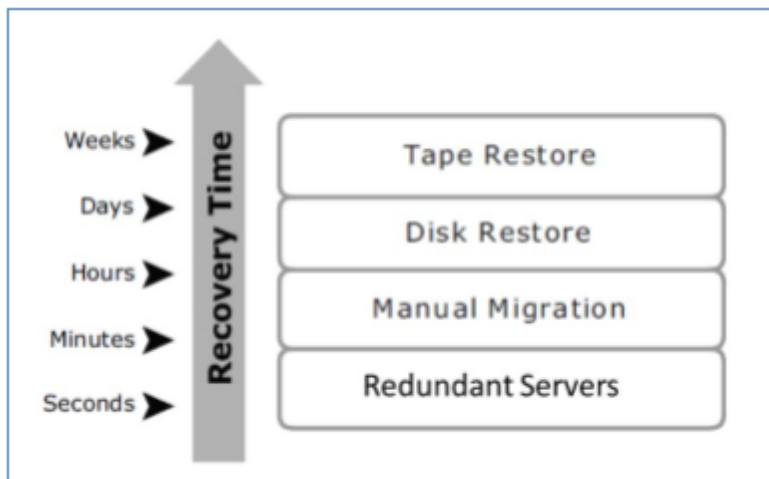
Fig.: RPO Strategy

Recovery-Time Objective (RTO): This refers to the time within which the system applications must be recovered in the event of an outage. It specifies the amount of downtime that a business can live with when a disaster occurs. After specifying the RTO for a particular data centre, the businesses can optimise its disaster recovery plans.

e.g. 1: If the RTO is 2 hours, and then instead of using tape backup, use disk backup, as a disk drive enables faster the recovery of data as compared to a tape drive.

e.g. 2: If the RTO is of one week, then tape backup can be used.

Figure shows an example of the RTO strategy. Here, the recovery strategy is defined based on the recovery time. If the RTO is in seconds or minutes, then redundant servers can be used. In case the RTO is hours, then manual data migration can be done. If the RTO is in days, then replicated data disk can be used. In case of RTO being in weeks, then tape can be used as means of data restoration.



RTO STRATEGY FIGURE

6. BC PLANNING LIFE CYCLE

Like any planning process, the BC planning should be done with a disciplined approach. Due to businesses using advanced computing technologies, there is a need within organisations to deploy dedicated resources for managing their BC operations. A BC life cycle consists of five stages as shown in Figure.



Fig. 5.1.4: BC Lifecycle Stages

Each stage is further divided into specific activities as listed below.

Stage 1 - Establishing objectives: In this stage, the objectives for BC are defined. These objectives indicate the kind of budget and skills required to implement the BC strategy. Listed below are the key activities of this stage:

- Finding out the requirements of BC
- Scoping and budgeting the estimation to meet the requirements
- Choosing a BC team, including domain experts from all business areas (external or internal)
- Generating the BC policies

Stage 2 - Analysing: This step is responsible for gathering the analytical information such as:

- Data profiles
 - Business processes
 - Infrastructure support
 - Dependencies
 - Frequency of using the business infrastructure
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- Defining the recovery priorities for critical business needs.
 - Performing risk analysis and defining the risk mitigation strategies.
 - Perform a business impact analysis (BIA) to understand the potential effects of failure on business operations.
 - Prepare the cost and benefit analysis. Examine the cost incurred in implementing a recovery strategy and weigh it against the benefit it would provide to a business in the event of a failure.
 - Weigh the options for implementing a data recovery solutions.

Stage 3 - Designing and developing: In this step, the data recovery solutions are initiated. The teams, security strategies and infrastructure for implementing BC are decided. Listed below are the key activities of this stage:

- Create a team and assign the roles and responsibilities to the team members. The team members would be responsible for the given below tasks:

- Emergency response
- Damage assessment
- Application recovery

- Design the data security strategies for protecting the data.

Develop the:

- Infrastructure for backup and recovery
- Contingency scenarios
- Emergency response measures
- Process for recovery and restart

Stage 4 - Implementing: Once the BC strategy is finalised, the next stage is about implementing the BC strategy. Listed below are the key activities of this stage:

- Implement the procedures for risk management and mitigation, such as backup and replication.
- Organise the failure recovery sites, which can be utilised if a failure affects the primary data centre.
- To avoid a single point of failure, introduce redundancy for each resource within the data centre.

Stage 5 - Training, testing, assessing and maintaining: BC strategies can change over time. To keep the BC implementation running efficiently, it is important to have an efficient team in place. Listed below are the key activities of this stage:

- Whenever the BC plan is modified, train the human resources who are responsible for backup and replication of business-critical data.
- Instruct the employees such that they respond spontaneously when the disasters

occur.

- Guide the recovery team on recovery measures based on emergency situations.
- Implement damage appraisal processes and evaluate recovery plan.
- Check the BC plan frequently to assess its performance and identify any limitations that need to be taken care of.
- Finalise the BC strategies and recovery processes frequently along with the changes in the data centre.

7. BC TECHNOLOGY SOLUTIONS

The BC technology solutions are required to overcome the failures after analysing the business impact of a system outage due to a failure. Once the impact of an outage on the business has been determined, the next stage is to define the appropriate recovery solutions. In order to recover the data and to resume the business operations, an alternate copy of the data is a must. The BC technology solutions define the strategy used for replicating the data.

The commonly used technology solutions for data replication include:

Backup and recovery: The most common method to take a backup is by using the tape drives. There are low-cost, high-capacity tape disks available in the market, which makes it an economical choice for backup. Further, tape drives provide high mobility, making it a good choice if tapes need to be shipped from one location to another.

The frequency for taking the backup of data is determined by the:

- RPO
- RTO
- Frequency of data changes

Storage array-based replication (local): In this strategy, the data is replicated to a different location, but within the same storage array. In case of a failure, the replicated storage array is used to access and restore data.

Storage array-based replication (remote): This strategy is similar to the local storage array based replication, with one difference. The storage array is located at a remote site and not locally. If there is a failure in the primary storage array, then the remote storage array can be used for data recovery and restoration.

Host-based replication: In this strategy, the replication operation is performed by the software that resides on application servers. The servers copy the data from one site to another. Though this is easy to implement, it can stress the server due to the additional responsibility of copying the data.

8. BACKUP AND RECOVERY CONSIDERATIONS

For any disaster recovery strategy to work, there are some backup and recovery considerations that need to be factored. Figure lists these considerations.

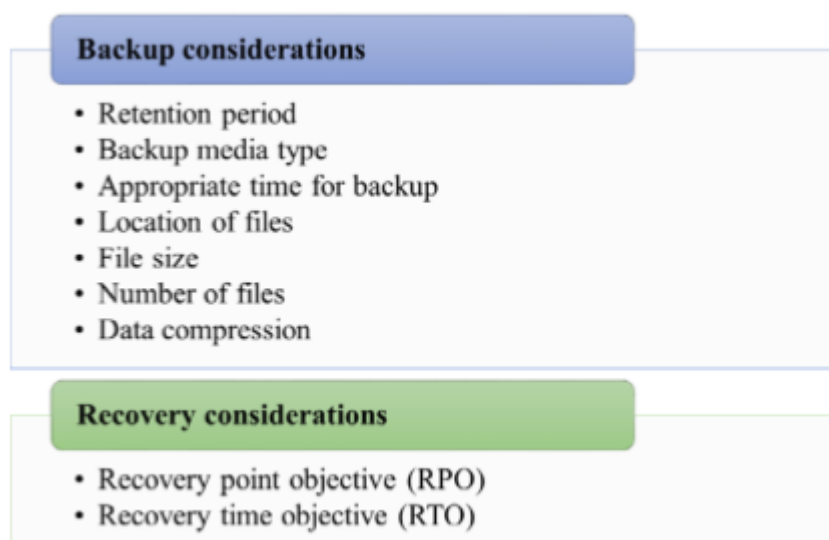


Figure: Backup and recovery considerations

Let us understand the factors that need to be considered for taking data backup:

- Retention period: This refers to the duration for which the backup copies need to be retained. Based on the data, the copies can be retained for a few days or for few years.
- Backup media type: This refers to the selection of the media type for taking the backup. The media type is chosen based on the retention period and the accessibility of data.
- Appropriate time for backup: This refers to the decision for selecting the most appropriate time for backup such that it does not disrupt the production operations.
- Location of files: Businesses can have data residing on multiple heterogeneous platforms. If the data centre uses backup data from many source, there may be issues of content integrity. Hence, it is important to define co-ordinated processes to backup data from multiple sources.
- File size: The amount of system resources that are used for the backup process depends on the file size. Taking a backup for a small amount of large-sized files would take less system resources as compared to taking a backup of equal amount of data for large number of small-sized files. For instance taking a backup of 5 files of 1 MB each consumes less system resources as compared to 5,000files of 1 KB each. If there are large number of small files, the backup and restore operations take more time.
- Number of files: The number of files that need to be backed up affects the efficiency of the backup process. Large number of files take longer time for backup and restoration as it takes more time to search for the data through the file system.
- Data compression: Data backup consumes large amount of media space. Therefore, the data compression is used in backup systems to save some media space. Some backup devices such as the tape drives have built-in hardware-based compression of data. Note that the effectiveness of data compression is determined by the type of data. For instance, text data may compress well, whereas application binaries may not compress well.

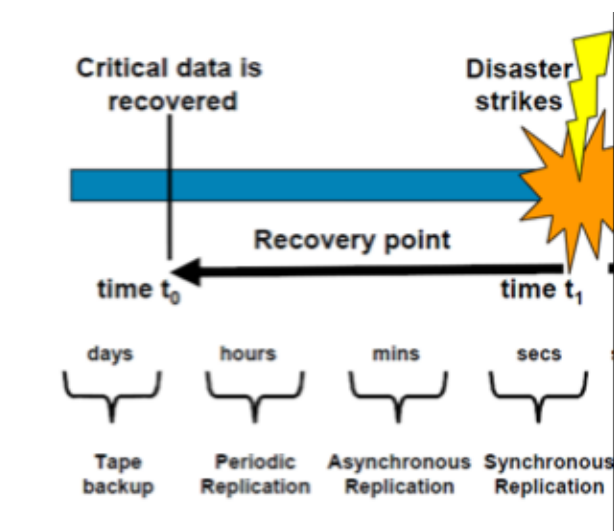
Let us understand the strategy that needs to be considered for data recovery:

- Recovery point objective (RPO): specifies the tolerable limit of data loss for a business and the time interval within which backups should be taken. e.g.: If the given RPO is for two hours, then the data need to be backed up is at least once in two hours.

Based on the RPO specified for operational recovery, the retention time for the backup is provided.

e.g.: 'X Company' may demand to re-establish the data from its operational backup copy, which is done a month ago. Based on operational recovery needs, 'X Company' can retain the data from one day to one month. If the given retention period is short, then it may be difficult to recover all the data for the requested recovery point (as it may be older than the retention period). When the given retention period is large, then it is possible to recover all the data to the requested recovery point, but this may require a large storage space and also its cost may increase. To overcome this limitation, it is necessary to describe the retention period based on the examination of all the restore requests done in the past and the budget available.

Figure depicts the RPO strategy from the fault occurrence to the data recovery phase. In this figure, the recovery point determines the recovery type. If the recovery period is within seconds, then the synchronous replication should be followed.



RECOVERY POINT OBJECTIVE

Recovery Time Objective (RTO): RTO refers to the time required for the recovery process to be completed. To meet the defined RTO, a combination of backup solutions can be used. This allows a business to minimise the recovery time. RTO also has an impact on the type of backup media used. For instance, if the data streams are multiplexed (multiple data streams written to a single tape drive

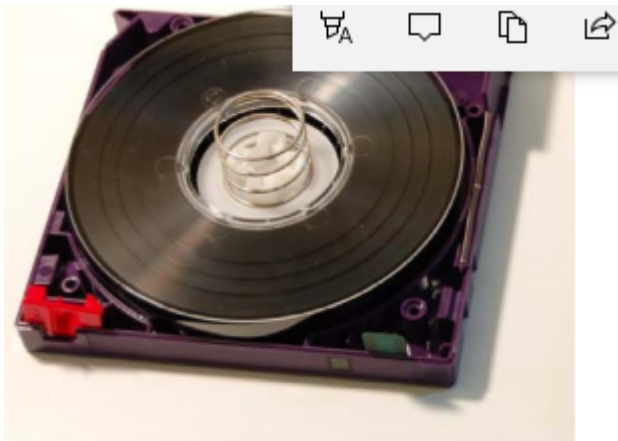
simultaneously) on tapes, then it would take more time for recovery as compared to tapes that do not have multiplexing. Figure depicts the RTO strategy from the fault occurrence to the data recovery phase. In this figure, the recovery point determines the recovery type. If the recovery time is in week, then restoration through tape drives can be used.

9. BACKUP TECHNOLOGIES

The two commonly technologies used for data backup are:

- Tapes
- Disks

Backup to Tape: Tape backup is a widely used low-cost technology. Tape drives are used to read/write the data from/to a tape cartridge. As the data can read or write consecutively, these tape drives are referred to linear devices. Figure shows a tape drive:



TAPE DRIVE

The method of inserting a tape cartridge into a tape drive is known as tape mounting. There are several types of tape cartridges based on the size, ability, nature, number of reels, solidity, tape length, tape width, tape tracks and speed. The tape cartridge is made of magnetic tape within a synthetic enclosed space with one or two reels.

Following are the recording methods for a tape drive:

- Linear recording method: An older technology in which data is written by multiple heads in parallel tracks. The recording spans across the whole tape.
- Helical scan method: In this method, the data is written on the tape diagonally.

- Linear serpentine method: Modern tape drivers use this method as it uses more tracks and fewer tape drive heads. The data written in this method is same as linear method, the only difference is that once the tape ends, the heads are moved and data recording happens backward.

Backup to Disk: Backup disks are increasingly being used for storing backup data instead tapes because of their performance advantages.

Following are some of the advantages of storing data to disk:

- Easier to implement
- Faster than tapes
- Less costly to implement
- Random access possible
- Redundant Array of Independent Disks (RAID) protection capabilities